

ANSIBLE

- **Ansible** is an open-source automation tool used for configuration management, application deployment, and infrastructure provisioning.
- It automates repetitive tasks across multiple servers.
- Ansible is **agentless**, meaning no software needs to be installed on managed nodes.
- It uses **SSH** to communicate with remote servers.

Uses of Ansible

- Server configuration management
- Application deployment
- Software installation
- Infrastructure provisioning
- Patch management
- Automation of repetitive tasks

Ansible Architecture

Control Node

- System where Ansible is installed.
- Executes playbooks and manages target servers.

Managed Nodes

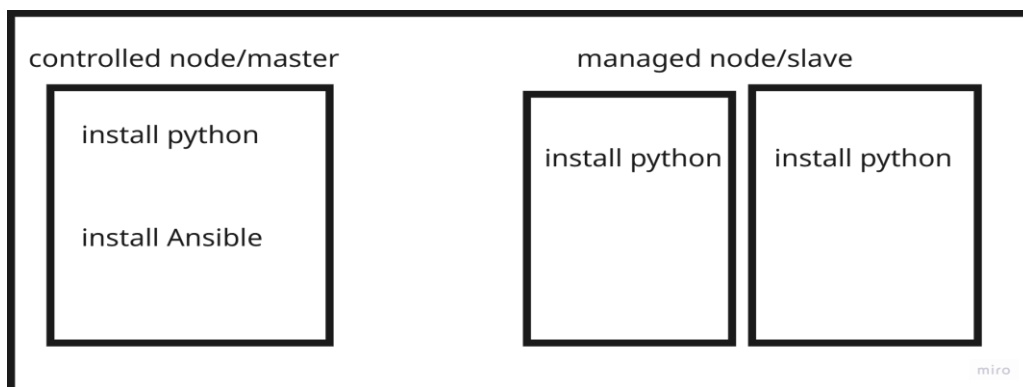
- Remote servers managed by Ansible.
- Connected through SSH.

Inventory File

- The inventory.txt file contains the private IP addresses of the Managed Nodes.
- Ansible uses these private IPs to connect and manage the remote servers.

Pageant

Pageant holds the private key (.ppk) of the instances and automatically provides it for SSH authentication.



HANDS-ON ANSIBLE

Screenshot: Create EC2 Instances for Ansible

- Open the AWS Management Console.
- Navigate to EC2 Dashboard.
- Launch 3 EC2 instances.
- Configure 1 instance as Control Node.
- Configure 2 instances as Managed Nodes.
- Select the same .ppk Key Pair for all instances.
- Instances created successfully.

The screenshot shows the AWS Management Console with the EC2 Instances page. Three instances are listed: 'managed_1' (t3.micro), 'managed_1' (t3.micro), and 'ansi-control' (t3.micro). All instances are in the 'Running' state. A 'Create key pair' dialog is open, showing the 'Key pair name' as 'ansi', the 'Key pair type' as 'RSA', and the 'Private key file format' as '.ppk'.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

ansi

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☐ .pem
For use with OpenSSH

☒ .ppk
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

Screenshot: Install Python and Ansible on Control Node

- Connect to the **Control Node** using SSH.
- Run `sudo yum update -y`.
- Run `sudo yum install python3 -y`.
- Run `sudo yum install ansible -y`.
- Verify the installation using `ansible --version`.

```
[ec2-user@ip-172-31-5-187 ~]$ sudo su
[root@ip-172-31-5-187 ec2-user]# #!/bin/bash
sudo yum update -y
sudo yum install python3-pip -y
sudo yum install ansible -y
```

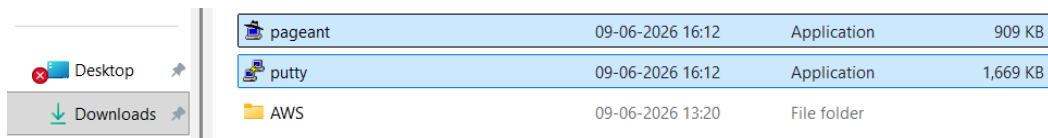
Screenshot: Install Python on Managed Nodes

- Connect to the **Managed Node 1** using SSH.
- Run `sudo yum update -y`.
- Run `sudo yum install python3 -y`.
- Repeat the same steps on **Managed Node 2**.

```
[ec2-user@ip-172-31-1-106 ~]$ sudo su
[root@ip-172-31-1-106 ec2-user]# #!/bin/bash
sudo yum update -y
sudo yum install python3-pip -y
```

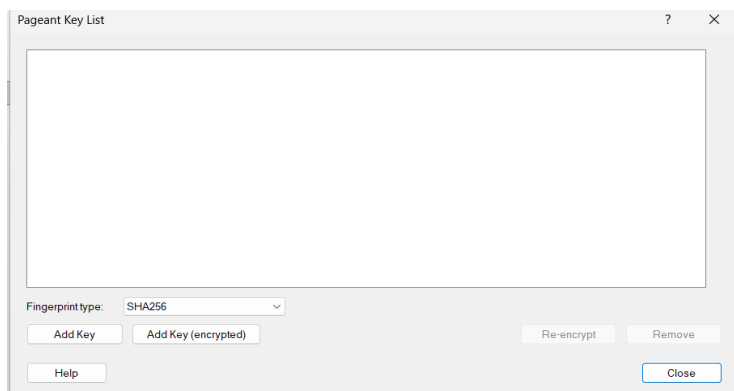
Screenshot: Download PuTTY and PuTTYgen

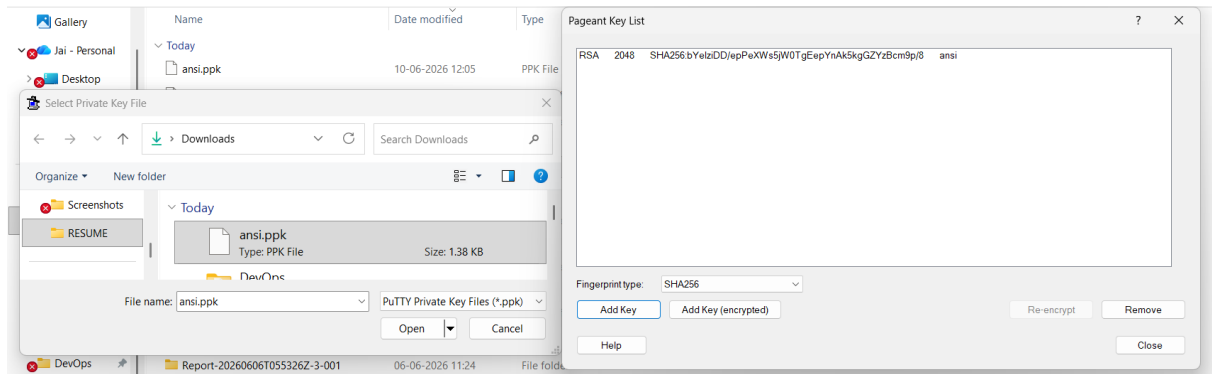
- Open a web browser.
- Download **PuTTY**.
- Download **PuTTYgen**.
- Install both applications successfully.



Screenshot: Configure PuTTY Pageant

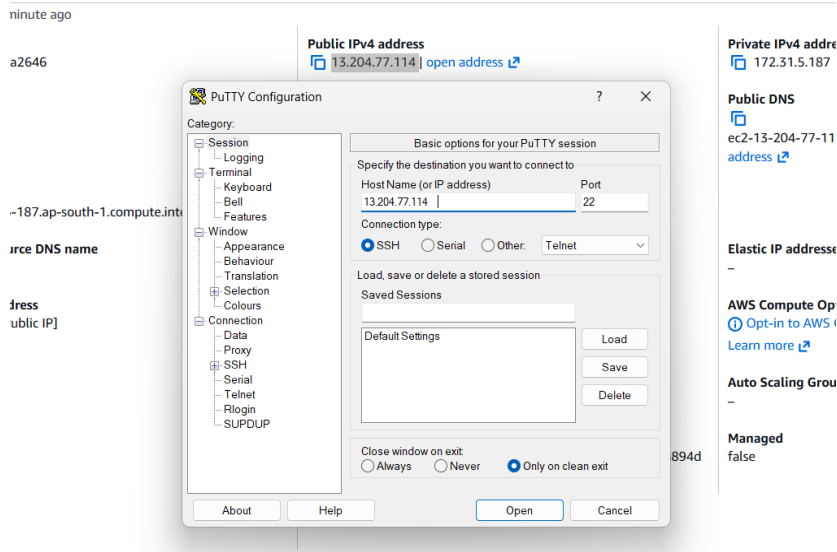
- Open Pageant.
- Right-click the Pageant icon in the system tray.
- Click Add Key.
- Select the .ppk file.
- Click Open.
- PPK key uploaded successfully.

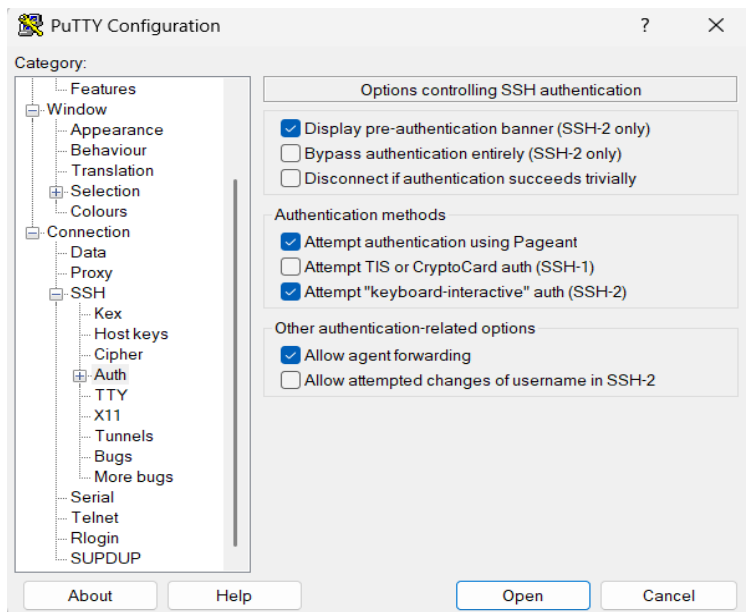




Screenshot: Connect to Control Node Using PuTTY

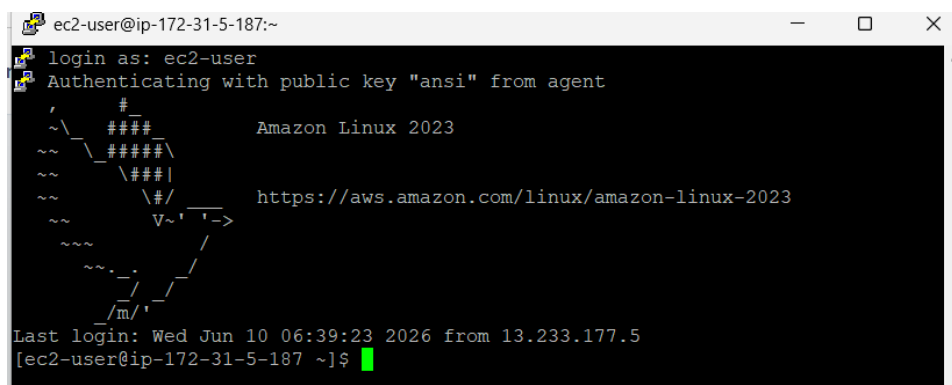
- Open **PuTTY**.
- Paste the **Control Node Public IP** in the **Host Name (or IP address)** field.
- In the left pane, navigate to **Connection** → **SSH** → **Auth**.
- Enable **Allow agent forwarding**.
- Click **Open**.
- Login to the Control Node successfully.





Screenshot: Login to PUTTY

- At the prompt login as:, enter ec2-user.
- Login to the Control Node successfully.



Screenshot: Create Inventory File

- Run vi inventory.txt.
- Paste the private IP addresses of both Managed Nodes.
- Press Esc.
- Type :wq!.
- Inventory file created successfully.

i-07b45197d0e772522 (managed_1)	i-098d885323d51c3b6 (managed_2)
PublicIPs: 43.204.142.125 PrivateIPs: 172.31.1.106	PublicIPs: 3.108.64.164 PrivateIPs: 172.31.7.254

```
ec2-user@ip-172-31-5-187 ~]$ vi inventory.txt
```

```
ec2-user@ip-172-31-5-187:~  
172.31.1.106  
172.31.7.254
```

```
[ec2-user@ip-172-31-5-187 ~]$ cat inventory.txt  
172.31.1.106  
172.31.7.254
```

Screenshot: Verify Connectivity to Managed Nodes

- Run `ansible all -i inventory.txt -m ping`.
- Press Enter.
- Ping response received from both Managed Nodes.
- Connectivity verified successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible all -i inventory.txt -m ping  
The authenticity of host '172.31.1.106 (172.31.1.106)' can't be established.  
ED25519 key fingerprint is SHA256:noHb8ZYLt+SoJHMTeommmvGBVZASWeftdcwUnjtr2dTE.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? [WARNING]: Platform linux on host 172.31.1.106 is using the discovered Python interpreter at /usr/bin/python3.9, but the installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.  
yes  
172.31.1.106 | SUCCESS => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3.9"  
  },  
  "changed": false,  
  "ping": "pong"  
}  
[ec2-user@ip-172-31-5-187 ~]$ ansible 172.31.7.254 -m ping  
172.31.7.254 | SUCCESS => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3.9"  
  },  
  "changed": false,  
  "ping": "pong"  
}
```

Screenshot: Install Apache on Managed Nodes

- Run `ansible web -i inventory.txt -m yum -a "name=httpd state=present" -b`.
- Press Enter.
- Apache (httpd) installed successfully on all Managed Nodes in the **web** group.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible all -i inventory.txt -m yum -a "name=httpd state=present" -b
[WARNING]: Platform linux on host 172.31.1.106 is using the discovered Python interpreter at /usr/bin/python3.9, but f
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html f
172.31.1.106 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": true,
  "msg": "",
  "rc": 0,
  "results": [
    "Installed: apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64",
```

Screenshot: Start Apache Service on Managed Nodes

- Run `ansible web -i inventory.txt -m service -a "name=httpd state=started" -b`.
- Apache (httpd) service started successfully on all Managed Nodes in the **web** group.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible all -i inventory.txt -m service -a "name=httpd state=started" -b
[WARNING]: Platform linux on host 172.31.1.106 is using the discovered Python interpreter at /usr/bin/python3.9,
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.
172.31.1.106 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": true,
  "name": "httpd",
  "state": "started",
  "status": {
    "AccessSELinuxContext": "system_u:object_r:httpd_unit_file_t:s0",
```

Screenshot: Access Apache Web Server

- Copy the **Public IP** of any one Managed Node.
- Open a web browser.
- Paste the Public IP in the address bar.
- The Apache (httpd) welcome page is displayed successfully.

i-07b45197d0e772522 (managed_1)

PublicIPs: **43.204.142.125** PrivateIPs: 172.31.1.106



Screenshot: Create Ansible Playbook

- Run `vi newpb.yaml`.
- Paste the Playbook configuration to uninstall Apache (httpd).
- Press Esc.
- Type `:wq!`.
- Playbook file created successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ vi newpb.yaml
[ec2-user@ip-172-31-5-187 ~]$
```

```
ec2-user@ip-172-31-5-187:~
- name: uninstall the httpd and then start
  hosts: all
  become: yes
  tasks:
    - name: uninstall the httpd
      yum:
        name: httpd
        state: absent
```

Screenshot: Run Ansible Playbook

- Run `ansible-playbook -i inventory.txt newpb.yaml`.
- Playbook executed successfully.
- Apache (httpd) uninstalled successfully on all Managed Nodes.

```
[ec2-user@ip-172-31-5-187 ~]$ vi newpb.yaml
[ec2-user@ip-172-31-5-187 ~]$ ansible-playbook -i inventory.txt newpb.yaml

PLAY [uninstall the httpd and then start] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.1.106 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of an
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.1.106]
[WARNING]: Platform linux on host 172.31.7.254 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of an
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.7.254]

TASK [uninstall the httpd] *****
changed: [172.31.1.106]
changed: [172.31.7.254]
```

Screenshot: Update Ansible Playbook

- Run `vi newpb.yaml`.
- Paste the Playbook configuration to install and start Apache (httpd) and copy the `index.html` file.
- Press `Esc`.
- Type `:wq!`.
- Playbook file updated successfully.

```
ec2-user@ip-172-31-5-187:~  
- name: install the httpd and then start  
  hosts: all  
  become: yes  
  tasks:  
    - name: install the httpd  
      yum:  
        name: httpd  
        state: present  
    - name: start the service  
      service:  
        name: httpd  
        state: started  
    - name: copy the index.html  
      copy:  
        src: index.html  
        dest: /var/www/html
```

Screenshot: Create Index File

- Run vi index.html.
- Paste the HTML code.
- Press Esc.
- Type :wq!.
- index.html file created successfully.

```
ec2-user@ip-172-31-5-187:~  
<!DOCTYPE html>  
<html>  
<head>  
<title>Page Title</title>  
</head>  
<body>  
  
<h1>This is a Heading</h1>  
<p>This is a paragraph.</p>  
  
</body>  
</html>
```

Screenshot: Run Ansible Playbook

- Run ansible-playbook -i inventory.txt newpb.yaml.
- Press Enter.
- Playbook executed successfully.
- Apache (httpd) installed and started successfully.
- index.html copied successfully to all Managed Nodes.

```
[ec2-user@ip-172-31-5-187 ~]$ vi newpb.yaml
[ec2-user@ip-172-31-5-187 ~]$ vi index.html
[ec2-user@ip-172-31-5-187 ~]$ vi newpb.yaml
[ec2-user@ip-172-31-5-187 ~]$
[ec2-user@ip-172-31-5-187 ~]$
[ec2-user@ip-172-31-5-187 ~]$ ansible-playbook -i in
index.html      inventory.txt
[ec2-user@ip-172-31-5-187 ~]$ ansible-playbook -i inventory.txt newpb.yaml

PLAY [install the httpd and then start] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.1.106 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.1.106]
[WARNING]: Platform linux on host 172.31.7.254 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.7.254]
```

Screenshot: Verify Web Page

- Copy the **Public IP** of any one Managed Node.
- Open a web browser.
- Paste the Public IP in the address bar.
- Press Enter.
- The custom index.html page is displayed successfully.



This is a Heading

This is a paragraph.

ANSIBLE ROLES

- **Ansible Roles** are used to organize Playbooks into reusable and structured components.
- Roles help manage complex automation by separating tasks, files, templates, variables, and handlers.
- They improve code reusability and maintainability.

Uses of Ansible Roles

- Organize automation code.
- Reuse configurations across multiple servers.
- Simplify large Playbooks.
- Manage applications and services efficiently.

HANDS-ON FOR ANSIBLE ROLE

Screenshot: Create Ansible Role

- Run `ansible-galaxy init myrole`.
- Press Enter.
- Ansible Role myrole created successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible-galaxy init myrole
- Role myrole was created successfully
[ec2-user@ip-172-31-5-187 ~]$ ls
index.html  inventory.txt  myrole  newpb.yaml
```

Screenshot: Install Tree Package

- Run `sudo yum install tree -y`.
- Press Enter.
- Tree package installed successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ sudo su
[root@ip-172-31-5-187 ec2-user]# yum install tree
Last metadata expiration check: 0:07:33 ago on Fri Jun 12 16:34:26 2026.
Dependencies resolved.
=====
Package      Architecture Version                                Repository      Size
=====
Installing:
tree         x86_64      1.8.0-6.amzn2023.0.2                  amazonlinux      56 k
Transaction Summary
=====
Install 1 Package
```

Screenshot: View Role Directory Structure

- Run `cd myrole`.
- Run `tree`.
- The Ansible Role directory structure is displayed successfully.

```
[root@ip-172-31-5-187 ec2-user]# cd myrole
[root@ip-172-31-5-187 myrole]# tree
.
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml

8 directories, 8 files
[root@ip-172-31-5-187 myrole]#
```

Screenshot: Create Role Playbook

- Run `vi role.yaml`.
- Paste the Playbook configuration.
- Press `Esc`.
- Type `:wq!`.
- `role.yaml` created successfully.

```
root@ip-172-31-15-214 ec2-user]# vi role.yaml
root@ip-172-31-15-214 ec2-user]#
```

```
root@ip-172-31-15-214:/home/ec2-user
- name: Install httpd on all servers
  hosts: all
  become: yes
  roles:
    - myrole
~
~
~
~
```

Screenshot: Create Role Tasks

- Run `cd myrole/tasks`.
- Run `vi main.yml`.

- Paste the task to install httpd.
- Press Esc.
- Type :wq!.
- main.yml updated successfully.

```
root@ip-172-31-15-214 ec2-user]# cd myrole/
root@ip-172-31-15-214 myrole]# ls
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars
root@ip-172-31-15-214 myrole]# cd tasks/
root@ip-172-31-15-214 tasks]# ls
main.yml
root@ip-172-31-15-214 tasks]# vi main.yml
root@ip-172-31-15-214 tasks]#
```

```
---
# tasks file for myrole

- name: install the httpd
  yum:
    name: httpd
    state: present
```

Screenshot: Run Role Playbook

- Run `ansible-playbook -i inventory.txt role.yaml`.
- Playbook executed successfully.
- Apache (httpd) installed successfully on all Managed Nodes using the Ansible Role.

```
bash: ansible-playbook: command not found
[root@ip-172-31-15-214 ec2-user]# ansible-playbook -i inventory.txt role.yaml

PLAY [Install httpd on all servers] *****

TASK [Gathering Facts] *****
The authenticity of host '172.31.11.133 (172.31.11.133)' can't be established.
ED25519 key fingerprint is SHA256:6PdLHSt15ubIn+A+uE0Yzbl9ODdjRcOE1AELS2pOdHo.
This key is not known by any other names
The authenticity of host '172.31.12.208 (172.31.12.208)' can't be established.
ED25519 key fingerprint is SHA256:enp6ZbPLDYa+kfgONIE2GrkgyFHoInsHqb9O8UndQ+E.
This key is not known by any other names
```

ENCRYPTION AND DECRYPTION

Screenshot: Encrypt Playbook File

- Run `ansible-vault encrypt newpb.yaml`.
- Enter the Vault password.
- Re-enter the Vault password for confirmation.
- Playbook file encrypted successfully.

```
[ec2-user@ip-172-31-5-187 ~]$
[ec2-user@ip-172-31-5-187 ~]$ ansible-vault encrypt newpb.yaml
New Vault password:
Confirm New Vault password:
Encryption successful
[ec2-user@ip-172-31-5-187 ~]$
```

Screenshot: Verify Encrypted Playbook

- Run cat newpb.yaml.
- Encrypted data is displayed successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ cat newpb.yaml
$ANSIBLE_VAULT;1.1;AES256
65363333366232363637383164323962363666393966613230626563333235346336306530626561
3430306236653536313161653334636666636236646638360a306132396162376535663562366330
6363363931633623837656236646663646165313363306461363565666334613864663531353536
3434326538303663660a333338653764393166646438336635343564303538613238373139656631
34643133363262386436376262353337636365633161303736616236326134363330326134326631
62643731336139363764396535666364643166326336633964393262313931663661353961326562
3737346566303626661376561306666393762333737616461633662306137613936383736343730
63656633313161323930356563343139666333643735646639366135666433393161326664653663
35323964323763663463366331323731346339373161626363633035336338626335353263366432
```

Screenshot: Create Encrypted File Using Ansible Vault

- Run ansible-vault create jai.yaml.
- Enter the Vault password.
- Re-enter the Vault password for confirmation.
- Type hello.
- Save and exit the file.
- Encrypted file jai.yaml created successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible-vault create jai.yaml
New Vault password:
Confirm New Vault password:
[ec2-user@ip-172-31-5-187 ~]$
```

```
hello
~
```

Screenshot: Verify Encrypted File

- Run cat jai.yaml.
- Encrypted text is displayed successfully.

```
[ec2-user@ip-172-31-5-187 ~]$
[ec2-user@ip-172-31-5-187 ~]$ cat jai.yaml
$ANSIBLE_VAULT;1.1;AES256
64653563316237623063306236613163376663633437393464636461323333376630306562633735
6538363164646463323930316364353365346161666262310a343836633733633135613335623235
65323866663466663661616234343565666165663365333864316164396261333839386337336637
3963316232343634380a643761653462353761643663653338613633383639306435373232313938
3638
[ec2-user@ip-172-31-5-187 ~]$
```

Screenshot: Decrypt Encrypted File

- Run `ansible-vault decrypt jai.yaml`.
- Enter the Vault password.
- File decrypted successfully.
- Run `cat jai.yaml`.
- The text `hello` is displayed successfully.

```
[ec2-user@ip-172-31-5-187 ~]$ ansible-vault decrypt jai.yaml
Vault password:
Decryption successful
[ec2-user@ip-172-31-5-187 ~]$ cat jai.yaml

hello
[ec2-user@ip-172-31-5-187 ~]$
```